Thus once the first spider is disturbed the process of sequential displacement from web to web may continue for several seconds, often causing a majority of the spiders in the aggregation to shift from their home refuge to an alien one' ('Social Spiders', *Scientific American*, March 1976). This is paradoxical in the sense of page 103.)*

What if individuals retain some memory of the outcome of past fights? This depends on whether the memory is specific or general. Crickets have a general memory of what happened in past fights. A cricket that has recently won a large number of fights becomes more hawkish. A cricket that has recently had a losing streak becomes more dovish. This was neatly shown by R. D. Alexander. He used a model cricket to beat up real crickets. After this treatment the real crickets became more likely to lose fights against other real crickets. Each cricket can be thought of as constantly updating his own estimate of his fighting ability, relative to that of an average individual in his population. If animals such as crickets, who work with a general memory of past fights, are kept together in a closed group for a time, a kind of dominance hierarchy is likely to develop.* An observer can rank the individuals in order. Individuals lower in the order tend to give in to individuals higher in the order. There is no need to suppose that the individuals recognize each other. All that happens is that individuals who are accustomed to winning become even more likely to win, while individuals who are accustomed to losing become steadily more likely to lose. Even if the individuals started by winning or losing entirely at random, they would tend to sort themselves out into a rank order. This incidentally has the effect that the number of serious fights in the group gradually dies down.

I have to use the phrase 'kind of dominance hierarchy', because many people reserve the term dominance hierarchy for cases in which individual recognition is involved. In these cases, memory of past fights is specific rather than general. Crickets do not recognize each other as individuals, but hens and monkeys do. If you are a monkey, a monkey who has beaten you in the past is likely to beat you in the future. The best strategy for an individual is to be relatively dovish towards an individual who has previously beaten him. If a batch of hens who have never met before are introduced to each other, there is usually a great deal of fighting. After a time the fighting dies down. Not for the same reason as in the crickets, though. In the case of the hens it is because each individual 'learns her place' relative to each other individual. This